INCH-POUND

MIL-C-18757/1B(OS)
24 January 1994
SUPERSEDING
MIL-C-18757/1A(OS)
27 November 1989

MILITARY SPECIFICATION

CARTRIDGE CASE, STEEL, 40-MILLIMETER, MK 3 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the 40-millimeter Mk 3 Mod 0 Steel Cartridge Case, herein referred to as the "cartridge case" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-D-3464	Desiccants Activated, Bagged, Packaging Use and Static Dehumidification
MIL-C-18757	Cartridge Cases, Steel (Other Than Small Arms), General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or sending a letter.

AMSC N/A FSC 1310

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STANDARDS

MILITARY

MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: Standardization Documents Order Desk , Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Codes 10001 and 53711)

330825	40mm Cartridge Case, Mk 3 Mod 0, Steel
258488	Receiver Gage, 40mm Cartridge Case
5871562	40mm Test Vehicle All-Up Round

(Application for copies of NAVSEA drawings should be addressed to: Commanding Officer, Naval Ordnance Station (Attn: Code 802), Louisville, KY 40214-5001.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

OD 47471	Method for Rating the Effects of Salt Spray Testing on Zinc Electroplated Cartridge Cases

(Application for copies should be addressed to: Commanding Officer, Naval Ship Weapon Systems Engineering Station, NAVSEA Data Support Activity (Code 5H30), Port Hueneme, CA 93043.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8 Tension Testing of Metallic Materials, Standard Test Methods of (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of the cartridge case shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 General requirements. The complete requirements for procuring the cartridge case described herein shall consist of MIL-C-18757, this document, and Drawing 330825.

3.3 Ammunition data card. Each production lot shall be accompanied by an ammunition data card prepared in accordance with MIL-STD-1167 (see 6.3).

3.4 Hardness. Hardness values and determinations shall fall within the ranges specified in table I and within the test regions as shown on figure 1 (see 4.5.1).

TABLE I. Hardness values [1/].

Region A	Region B	Region C		Region D		Region E		Region F	
Max	Min	Max	Min	Max	Min	Max	Min	Max	Min
90B	90B	38C	95B	38C	30C	38C	90B	89B	75B

[1/] All values given in Rockwell scale.

3.4.1 Cartridge case rim. The rim of the cartridge case (region A) shall be annealed by a suitable method to reduce the hardness and meet the values shown in table I in order to prevent excessive gun wear. The area affected by this annealing treatment shall not extend beyond the limits of region A.

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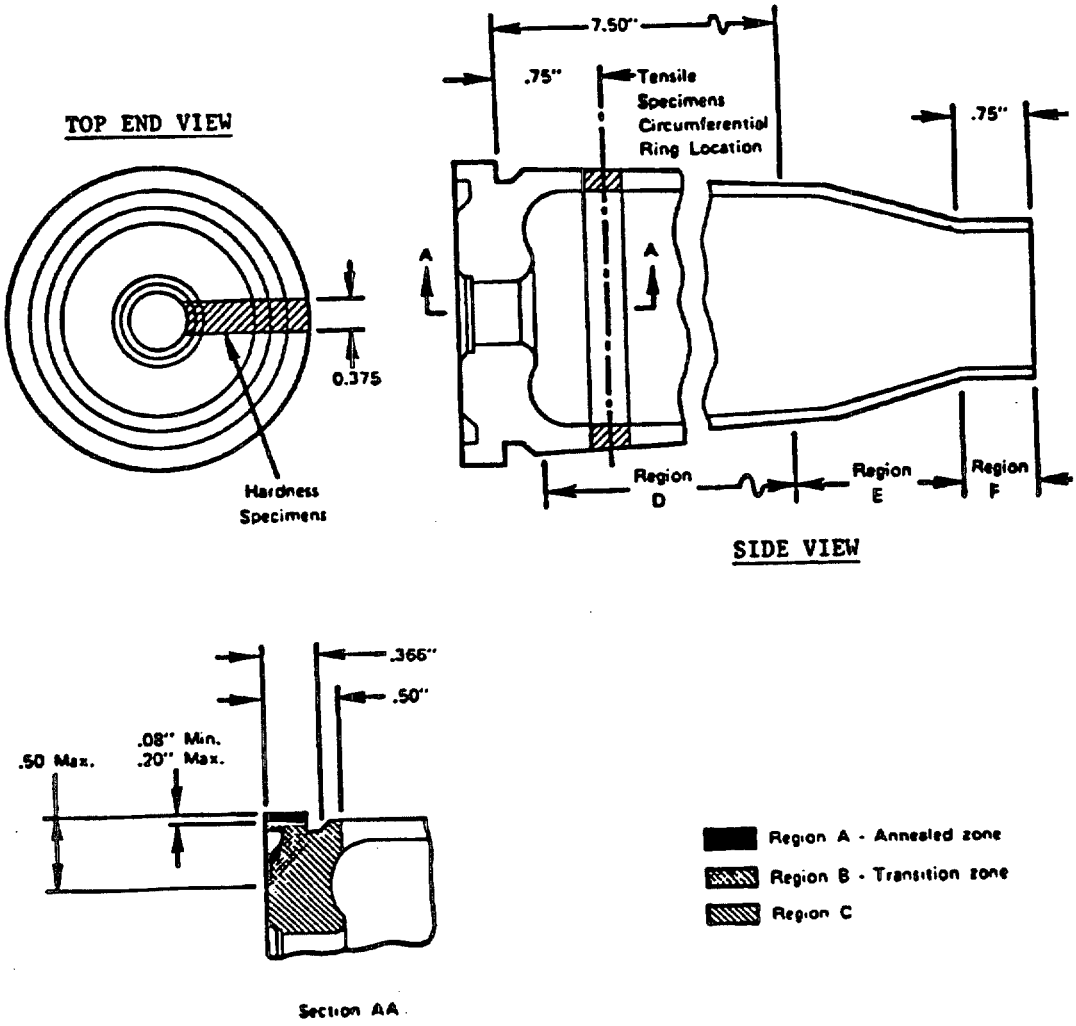


FIGURE 1. Hardness and tensile specimen locations.

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3.5 Elongation and tension. The elongation values and yield strength values shall be as specified in table II at the location shown on figure 1 (see 4.5.2 and 4.5.3).

TABLE II. Yield strength and elongation.

Region	Yield Strength	Elongation in 1 Inch
D	(Offset - 0.2 percent) 115,000 psi	6.05 percent

3.6 Protective coating. The cartridge case protective coating shall meet the requirements specified on Drawing 330825 and shall have a rating of the black corrosion products of zinc less than the established standard of 0.2400 when tested in accordance with 4.5.4.

3.7 Ballistic test. Each cartridge case shall load, extract, and eject without difficulty when ballistically tested in accordance with 4.5.5. Each cartridge case shall be from from cracks, tears, or splits and separations after ballistic testing.

3.8 Workmanship. Workmanship shall be such that the cartridge cases are uniform in quality and condition, free from cracks, tears, or splits and separations, foreign materials, and surface imperfections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Production lot acceptance inspection (see 4.4)

4.3 First article inspection. The first article sample shall consist of 158 cartridge cases and shall be examined and tested in accordance with table I of MIL-C-18757 and table III of this specification. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

TABLE III. Sample size for specific tests.

Type of Inspection	Total sample size	Ballistic sample	Tensile and Hardness	Protective Coating
First article	158	150	5[1/]	3
Production	58	50	5[1/]	3

[1/] Exception to note 1 to table I, MIL-C-18757: Hardness readings shall be obtained for all regions listed in table I of this specification from number of samples indicated.

4.4 Production lot acceptance inspection. Unless otherwise specified in the contract (see 6.2), the production lot shall consist of 5,000 cartridge cases plus those required for test purposes and shall be examined and tested in accordance with table I of MIL-C-18757 and table III of this specification. After production of the first ten lots on any contract, successive lot size may be increased to 20,000 by the contracting activity, with test quantities to remain the same as those required for lots of 5,000.

4.4.1 Reinspection and retest. Reinspections and retests of any lot shall be as directed by the contracting activity.

4.5 Tests.

4.5.1 Hardness. On cartridge cases selected for hardness and tensile tests, circumferential ring for the tension test will first be cut from cartridge cases as shown on figure 1. After this, a longitudinal section with a nominal width of 0.375 inch will be cut from the remaining parts (see figure 1, end view) of the cartridge case to be used for obtaining hardness values for all regions specified in table I. On the circumferential ring for tension test, obtain hardness readings by taking measurements in the specimen grip area or adjacent to the grip area away from the necked down area. Record all measurements. Tests shall be in accordance with the hardness test of MIL-C-18757.

4.5.2 Elongation. Cartridge cases shall be prepared and tested in accordance with the elongation test of MIL-C-18757.

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4.5.3 Tension. Cut and identify circumferential rings from sample cases. Maintain ring widths slightly greater than the final specimen grip width to allow for subsequent shaping operations. Prior to cutting the rings, some surface machining (both inside and outside surfaces) may be required to produce parallel specimen face surfaces (nominal thickness to be between 0.050 and 0.070 inch). Remove edge burrs and sand edges slightly to remove any possible stress raisers. Straighten specimens by passing them through a sheet metal roller, using a minimum of passes, until bowing cannot be detected with a straight edge. Machine all specimen segments to dimensions specified in ASTM E 8, Figure 1, Subsize Specimens. Several specimens may be machined at one time. Individually, sand the edges of the gage section to obtain a 0.003- to 0.005-inch taper from the end to the middle of the gage section. Smooth all edges of the necked down area. Stress relieve all segments at 321 ± 5°C for 30 ± 5 minutes and then cool to room temperature for a minimum of 1 hour. Test all tensile specimens using a head separation rate of 0.05 inch per minute. Retain original stress-strain curves. Note magnification and scale on all stress-strain curves. Measure gage dimensions as accurately as possible. Record gage cross sectional area to four decimal places.

4.5.4 Protective coating. Cartridge cases shall be prepared and tested in accordance with the protective coating test of MIL-C-18757. The assigned plating thickness value of the interior recessed area of the cartridge case head shall be determined as follows: Three specimen sections shall be taken from each of three cases, approximately 120 degrees apart, and six measurements, approximately 0.1 inch apart, shall be taken on each segment (see figure 2). The assigned plating thickness value (T) shall be calculated as follows:

T = plating thickness, any case
t_a = plating thickness, segment a
t_b = plating thickness, segment b
t_c = plating thickness, segment c
t_{a1} = plating thickness reading preceding t
t_{a2} = minimum plating thickness reading of segment a
t_{a3} = plating thickness reading following t_{a2}

$$t_a = \frac{t_{a1} + 6\,t_{a2} + t_{a3}}{3}$$

$$T = \frac{t_a + t_b + t_c}{3}$$

t_b and t_c are to be completed in a similar manner.

NOTES: (1) If t_{a2} is a measurement at location 1 or 6, use $t_{a1} = t_{a3}$ in above formula (use on any segment). See figure 2.

(2) Record all thickness readings. The black speck corrosion rating procedure shall be as specified in OD 47471.

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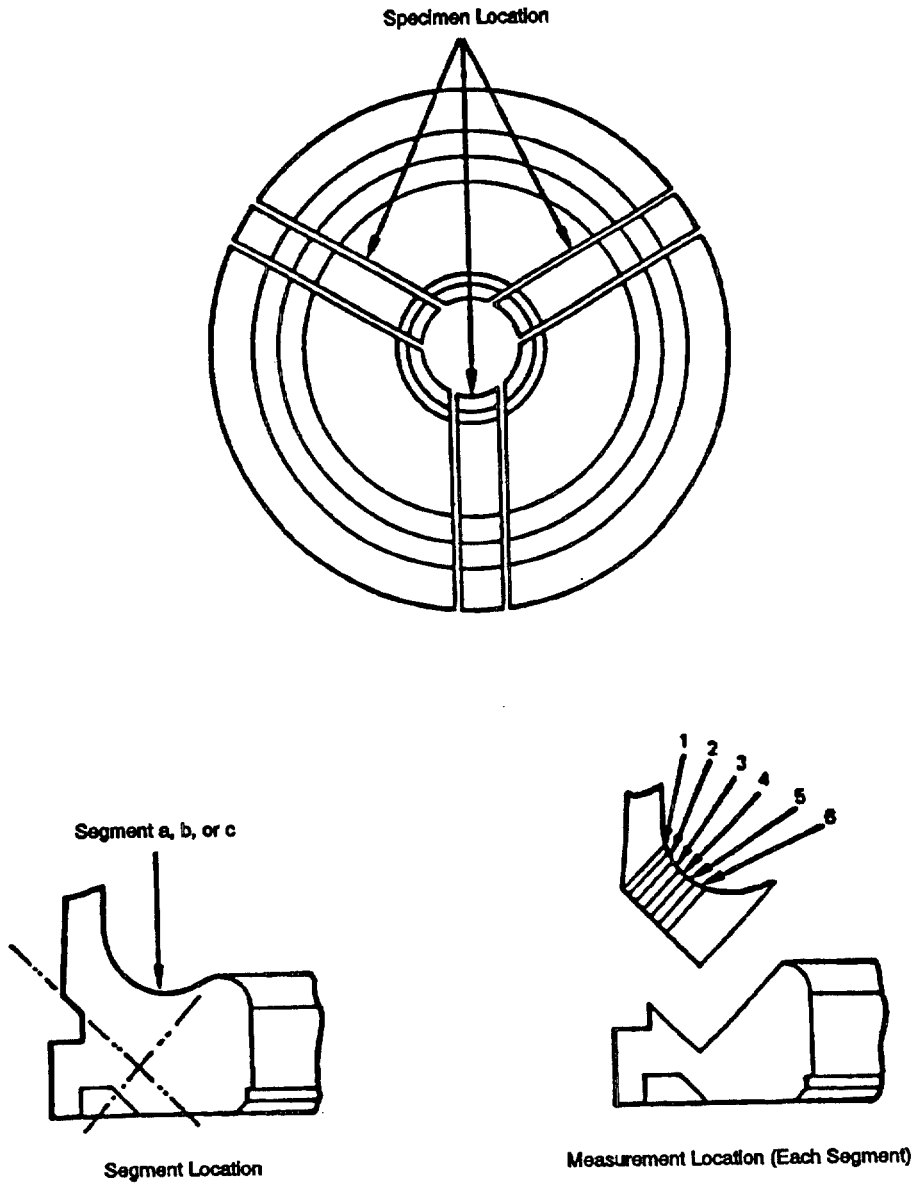


FIGURE 2. Metallographic sectioning.

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4.5.5 Ballistic test. The quantity of cartridge cases for ballistic testing shall be as specified in table III. The cartridges shall be loaded in accordance with Drawing 5871562 with an adjusted propellant charge weight to produce the pressure specified in 4.5.5.1.1. Firing shall be in accordance with table IV.

TABLE IV. Test firings.

Type of lot	Number of rounds	Charge pressure	Round temperature °C
First article	70	Service	21 to 35
	30	Proof	21 to 35
	25	Service	49
	25	Adjusted to give service pressure	-34.4
Production	40	Service	
Lot acceptance	10	Proof	

4.5.5.1 Proof pressure. Each round for testing at proof pressure shall be fired once in a single shot, unless otherwise directed by the contracting activity. For service pressure test, fire five rounds in single shot action with pressure gages, to determine compliance with pressure requirements, and fire the remainder of rounds in semi-automatic rapid fire action, using automatic loaders, in five round bursts. Temperature of first article rounds shall be as specified in table IV. Temperature of production rounds shall be as directed by the contracting activity. Rounds shall be temperature conditioned a minimum of 1 hour more than the time necessary to bring them to the specified temperature.

4.5.5.1.1 Ballistic test pressure. The propellant charge shall be adjusted so that the ballistic test pressure at any temperature is as follows:

Service charge: 19.0 to 20.0 tons
Proof charge: 21.6 to 23.6 tons

Pressure measurements are in long tons as determined by copper crusher gages. One long ton = 2240 pounds (US).

These charges should not be confused with the charges used in the testing of propellants, and apply only to the testing of cartridge cases.

4.5.6 Chamber gage examination. Each cartridge case shall be chamber gaged in receiver, Drawing 258488, to determine acceptability. A knockout force of up to 15 pounds may be used to remove case from chamber gage.

4.6 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of 5.1 and 5.2.

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5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with MIL-C-18757.

5.1.1 Packing. Packing shall be in accordance with MIL-C-18757. Two units of desiccant, MIL-B-3464, Type I, shall be added to packing box before sealing.

5.2 Marking. Unless otherwise specified by the contracting activity, marking of the lot numbers for each ammunition lot shall be in accordance with MIL-STD-1168.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 40-mm Mk 3 Mod 0 steel cartridge cases, when assembled with their appropriate primers, propellant charges, and loaded projectiles, form complete rounds which are used as service ammunition in the 40-mm anti-aircraft missile.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)

c. Whether a first article is required (see 3.1 and 6.4)

d. Agency responsible for inspection if different from 4.1

e. Production lot size, if other than as specified in 4.5.1

f. Proof pressure test method if different from 4.5.5.1

g. Temperature of production rounds in proof pressure testing of 4.5.5.1

h. Lot numbering, if other than as specified in 5.2

i. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material," DOD 4145.26M, are applicable (see 6.4).

6.3 Data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct

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contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Ref. Paragraph	DID Number	DID Title	Tailoring
3.3	DI-MISC-80043	Ammunition Data Card	—

(The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.)

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Federal Acquisition Regulations (FAR) 23.3.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.6 Subject term (key word) listing.

Ammunition cases
Arms, small
Ballistic tests

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
Navy - OS
(Project 1310-N556)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-18757/1B(OS)	1994 JANUARY 24

3. DOCUMENT TITLE
CARTRIDGE CASE, STEEL, 40-MILLIMETER, MK 3 MOD 0

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER INDIAN HEAD DIVISION, NSWC	(1) Commercial 301-743-4510/4253	(2) AUTOVON 354-4510/4253
c. ADDRESS (Include Zip Code) 101 STRAUSS AVE. CODE 8420 INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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